

COURSE MATERIAL

SUBJECT	BASIC ELECTRICAL & ELECTRONICS ENGINEERING(EE23AES101)
UNIT	1
COURSE	B.TECH
DEPARTMENT	ECE
SEMESTER	1-1
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1. Course Objectives

The objectives of the course are to make the students learn about:

1. To introduce basics of electric circuits.
2. To teach DC and AC electrical circuit analysis.
3. To explain working principles of transformers and electrical machines.
4. To impart knowledge on low voltage electrical installations

2. Prerequisites

Students should have knowledge on

1. Mathematics.
2. Physics.

3. Syllabus

UNIT-I

DC & AC Circuits:

Electrical circuit elements (R - L and C) - Kirchhoff laws - Series and parallel connection of resistances with DC excitation. Nodal and Mesh analysis. Superposition Theorem - Representation of sinusoidal waveforms - average and rms values - phasor representation - real power - reactive power - apparent power - power factor - Analysis of single-phase ac circuits consisting of RL - RC - RLC series circuits.

4. Course outcomes

1. **Recall** Kirchhoff laws (L2).
2. **Analyze** simple electric circuits with DC excitation (L4).
3. **Apply** network theorems to simple circuits (L3).
4. **Analyze** single phase AC circuits consisting of series RL - RC - RLC combinations (L4).

5. Co-PO / PSO Mapping

Machine Tools	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	P10	PO11	PO12	PSO1	PSO2
CO1	3	3												
CO2	3	3												
CO3	3	3												
CO4	3	3												

6. Lesson Plan

Lecture No.	Weeks	Topics to be covered	References
1	1	Introduction	T1, R1
2		Electrical circuit elements (R - L and C)	T1, R1
3		Kirchhoff laws	T1, R1
4		Series and parallel connection of resistances with DC excitation	T1, R1
5		Nodal analysis	T1, R2
6		Mesh analysis	T1, R1
7	2	Superposition Theorem	T1, R1
8		Representation of sinusoidal waveforms	T1, R1
9		Average and rms values - phasor representation	T1, R1
10		Real power - reactive power - apparent power - power factor	T1, R1
11	3	Analysis of single-phase ac circuits consisting of RL - RC series circuits.	T1, R1
12		Analysis of single-phase ac circuits consisting of RLC series circuits.	T1, R1
13		Numerical Problems	T1, R1
14		Numerical Problems	T1, R1

7. Activity Based Learning

1. Explaining circuits in laboratory for practical knowledge.
2. Mini project on designing basic circuits.

8. Lecture Notes

1.1 INTRODUCTION

An Electric circuit is an interconnection of various elements in which there is at least one closed path in which current can flow. An Electric circuit is used as a component for any engineering system.

The performance of any electrical device or machine is always studied by drawing its electrical equivalent circuit. By simulating an electric circuit, any type of system can be studied for e.g., mechanical, hydraulic thermal, nuclear, traffic flow, weather prediction etc.

All control systems are studied by representing them in the form of electric circuits. The analysis, of any system can be learnt by mastering the techniques of circuit theory.

The analysis of any system can be learnt by mastering the techniques of circuit theory.

1.2 ELEMENTS OF AN ELECTRIC CIRCUIT

An Electric circuit consists of following types of elements.

1.2.1 Active elements:

Active elements are the elements of a circuit which possess energy of their own and can impart it to other element of the circuit.

Active elements are of two types

- a) Voltage source
- b) Current source

A Voltage source has a specified voltage across its terminals, independent of current flowing through it.

A current source has a specified current through it independent of the voltage appearing across it.

1.2.2 Passive Elements:

The passive elements of an electric circuit do not possess energy of their own. They receive energy from the sources. The passive elements are the resistance, the inductance and the capacitance. When electrical energy is supplied to a circuit element, it will respond in one and more of the following ways.

If the energy is consumed, then the circuit element is a pure resistor.

If the energy is stored in a magnetic field, the element is a pure inductor.

And if the energy is stored in an electric field, the element is a pure capacitor.

1.2.3 Linear and Non-Linear Elements.

Linear elements show the linear characteristics of voltage & current. That is its voltage-current characteristics are at all-times a straight-line through the origin.

For example, the current passing through a resistor is proportional to the voltage applied through its and the relation is expressed as V is directly proportional to I or $V = IR$. A linear element or network is one which satisfies the principle of superposition, i.e., the principle of homogeneity and additivity.

Resistors, inductors and capacitors are the examples of the linear elements

and their properties do not change with a change in the applied voltage and the circuit current.

Non linear element's V-I characteristics do not follow the linear pattern i.e. the current passing through it does not change linearly with the linear change in the voltage across it. Examples are the semiconductor devices such as diode, transistor.

1.2.4 Bilateral and Unilateral Elements:

An element is said to be bilateral, when the same relation exists between voltage and current for the current flowing in both directions.

Ex: Voltage source, Current source, resistance, inductance & capacitance. The circuits containing them are called bilateral circuits.

An element is said to be unilateral, when the same relation does not exist between voltage and current when current flowing in both directions. The circuits containing them are called unilateral circuits.

Ex: Vacuum diodes, Silicon Diodes, Selenium Rectifiers etc.

1.3 R L C PARAMETERS:

1.3.1 Resistance:

Resistance is that property of a circuit element which opposes the flow of electric current and in doing so converts electrical energy into heat energy.

It is the proportionality factor in ohm's law relating voltage and current.

Ohm's law states that the voltage drop across a conductor of given length and area of cross section is directly proportional to the current flowing through it.

Where the reciprocal of resistance is called conductance G . The unit of resistance is ohm and the unit of conductance is mho or Siemens.

When current flows through any resistive material, heat is generated by the collision of electrons with other atomic particles. The power absorbed by the resistor is converted to heat and is given by the expression

$$P = vi = i^2R$$

where i is the resistor in amps, and v is the voltage across the resistor in volts.

1.3.2 Inductance:

change in the electromagnetic field. This induces a voltage across the coil according to Faradays laws of Electromagnetic Induction.

$$\text{Induced Voltage } V = L \frac{di}{dt}$$

V = Voltage across inductor in volts

I = Current through inductor in amps

$$di = \frac{1}{L} v dt$$

Integrating both sides,

$$\int_0^t di = \frac{1}{L} \int_0^t v dt$$

$$\text{Power absorbed by the inductor } P = VI = Li \frac{di}{dt}$$

Energy stored by the inductor

$$W = \int_0^t P dt = \int_0^t Li^2 \frac{di}{dt} dt = \frac{Li^2}{2}$$

$$W = \frac{Li^2}{2}$$

Conclusions:

$$1) \quad V = L \frac{di}{dt}$$

The induced voltage across an inductor is zero if the current through it is constant. That means an inductor acts as short circuit to dc.

- 2) For minute change in current within zero time ($dt = 0$) gives an infinite voltage across the inductor which is physically not at all feasible. In an inductor, the current cannot change abruptly. An inductor behaves as open circuit just after switching across dc voltage.
- 3) The inductor can store finite amount of energy, even if the voltage across the inductor is zero.
- 4) A pure inductor never dissipates energy, it only stores it. Hence it is also called as a non-dissipative passive element. However, physical inductor dissipates power due to internal resistance.

Ex: The current in a 2H inductor raises at a rate of 2A/s. Find the voltage across the inductor the energy stored in the magnetic field at after 2sec.

Sol:

$$V = L \frac{di}{dt}$$

$$= 2 \times 2 = 4V$$

$$W = \frac{Li^2}{2} = \frac{2 \times (4)^2}{2} = 16 J$$

1.3.3 Capacitance:

- A capacitor consists of two metallic surfaces or conducting surfaces separated by a dielectric medium.
- It is a circuit element which is capable of storing electrical energy in its electric field.
- Capacitance is its capacity to store electrical energy.
- Capacitance is the proportionality constant relating the charge on the conducting plates to the potential.

Charge on the capacitor q directly proportional to V

$$q = CV$$

Where `C` is the capacitance in farads, if q is charge in coulombs and V is the potential difference across the capacitor in volts.

The current flowing in the circuit is rate of flow of charge

$$i = \frac{dq}{dt} = C \frac{dv}{dt}$$

The capacitance of a capacitor depends on the dielectric medium & the physical dimensions. For a parallel plate capacitor, the capacitance

$$C = \frac{\epsilon A}{D} = \epsilon_0 \epsilon_r \frac{A}{D}$$

A is the surface area of plates

D is the separation between plates

ϵ is the absolute permeability of medium

ϵ_0 is the absolute permeability of freespace

ϵ_r is the relative permeability of medium

$$i = \frac{dq}{dt} = C \frac{dv}{dt}$$

$$\frac{dv}{dt} = \frac{i}{C}$$

$$V = \frac{1}{C} \int i dt$$

The power absorbed by the capacitor $P = v i = v C \frac{dv}{dt}$

$$\begin{aligned} \text{Energy stored in the capacitor } W &= \int_0^t P dt = \int_0^t V C \frac{dv}{dt} dt \\ &= C \int_0^t v dv = \frac{1}{2} C v^2 \text{ Joules} \end{aligned}$$

This energy is stored in the electric field set up by the voltage across capacitor.

Conclusions:

1. The current in a capacitor is zero, if the voltage across it is constant, that means the capacitor acts as an open circuit to dc
2. A small change in voltage across a capacitance within zero time gives an infinite current through the capacitor, which is physically impossible.
 - ✱ In a fixed capacitor, the voltage cannot change abruptly
 - ✱ A capacitor behaves as short circuit just after switching across dc voltage.
3. The capacitor can store a finite amount of energy, even if the current through it is zero.
4. A pure capacitor never dissipates energy but only stores it hence it is called non-dissipative element.

1.4 KIRCHHOFF'S LAWS:

Kirchhoff's laws are more comprehensive than Ohm's law and are used for solving electrical networks which may not be readily solved by the latter.

Kirchhoff's laws, two in number, are particularly useful in determining the equivalent resistance of a complicated network of conductors and for calculating the currents flowing in the various conductors.

1.4.1 Kirchhoff's Current Law (KCL)

In any electrical network, the algebraic sum of the currents meeting at a point (or junction) is Zero.

That is the total current entering a junction is equal to the total current

leaving that junction. Consider the case of a network shown in Fig (a).

$$I_1 + (-I_2) + (I_3) + (+I_4) + (-I_5) = 0$$

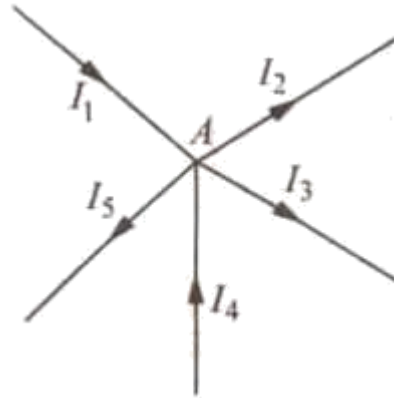
$$I_1 + I_4 - I_2 - I_3 - I_5 = 0$$

Or

$$I_1 + I_4 = I_2 + I_3 + I_5$$

Or

Incoming currents = Outgoing currents



1.4.2 Kirchhoff's Mesh Law or Voltage Law (KVL)

In any electrical network, the algebraic sum of the products of currents and resistances in each of the conductors in any closed path (or mesh) in a network plus the algebraic sum of the e.m.f.'s in that path is zero.

That is, $\sum IR + \sum \text{e.m.f} = 0$ round a mesh

It should be noted that algebraic sum is the sum which takes into account the polarities of the voltage drops.

That is, if we start from a particular junction and go round the mesh till we come back to the starting point, then we must be at the same potential with which we started.

Hence, it means that all the sources of emf met on the way must necessarily be equal to the voltage drops in the resistances, every voltage being given its proper sign, plus or minus.

Determination of Voltage Sign

In applying Kirchhoff's laws to specific problems, particular attention should be paid to the algebraic signs of voltage drops and e.m.f.s.

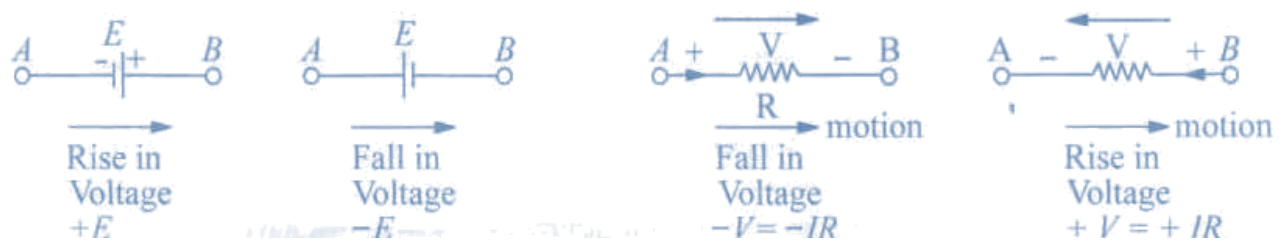
Sign of Battery E.M.F.

A rise in voltage should be given a + ve sign and a fall in voltage a -ve sign.

That is, if we go from the -ve terminal of a battery to its +ve terminal there is a rise in potential, hence this voltage should be given a +ve sign.

And on the other hand, we go from +ve terminal to -ve terminal, then there is a fall in potential, hence this voltage should be preceded by a -ve sign.

The sign of the battery e.m.f is independent of the direction of the current through that branch.

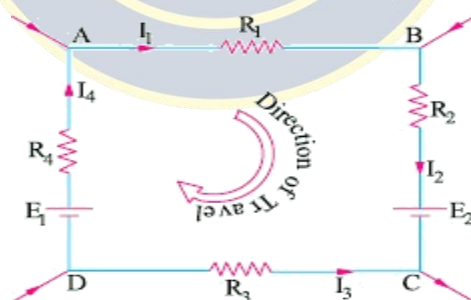


Sign of IR Drop

Now, take the case of a resistor (Fig.). If we go through a resistor in the same direction as the current, then there is a fall in potential because current flows from a higher to a lower potential..

Hence, this voltage fall should be taken -ve. However, if we go in a direction opposite to that of the current, then there is a rise in voltage. Hence, this voltage rise should be given a positive sign.

Consider the closed path ABCDA in Fig.



As we travel around the mesh in the clockwise direction, different voltage drops will have the following signs :

Using Kirchhoff's voltage law, we get

$$-I_1 R_1 - I_2 R_2 - I_3 R_3 - I_4 R_4 - E_2 + E_1 = 0$$

Or

$$I_1 R_1 + I_2 R_2 + I_3 R_3 + I_4 R_4 = E_1 - E_2$$

Assumed Direction of Current:

In applying Kirchhoff's laws to electrical networks, the direction of current flow may be assumed either clockwise or anticlockwise. If the assumed direction of current is not the actual direction, then on solving the question, the current will be found to have a minus sign.

If the answer is positive, then assumed direction is the same as actual direction. However, the important point is that once a particular direction has been assumed, the same should be used throughout the solution of the question.

Kirchhoff's laws are applicable both to d.c. and a.c. voltages and currents. However, in the case of alternating currents and voltages, any e.m.f. of self-inductance or that existing across a capacitor should be also taken into account.

1.5 NETWORK REDUCTION TECHNIQUES:

1.5.1 Series Connection of Resistors:

Two or more resistors in a circuit are said to be in series when the current flowing through all the resistors is the same.



Consider the circuit in fig(a), where two resistors R_1 and R_2 are in series, since the same current i flows in both of them. Applying Ohm's law to each of the resistors, we obtain

$$v_1 = iR_1, \dots\dots (1) \quad v_2 = iR_2 \quad (2)$$

If we apply KVL to the loop fig(b), we have

$$-v + v_1 + v_2 = 0 \quad (3)$$

$$v = v_1 + v_2 = i(R_1 + R_2)$$

$$i = v / (R_1 + R_2) \quad (4)$$

$$v = iR_{eq} \quad (5)$$

Implying that the two resistors can be replaced by an equivalent resistor; that is,

$$R_{eq} = R_1 + R_2$$

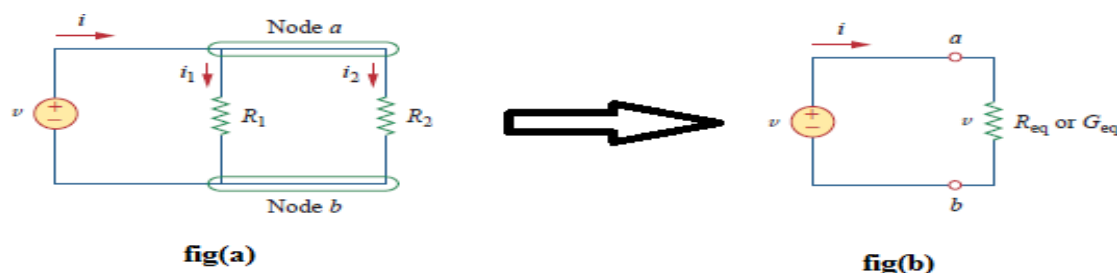
Note: The equivalent resistance of any number of resistors connected in series is the sum of the individual resistances.

If "n" resistors are in series, $R_{eq} = R_1 + R_2 + \dots + R_n$

If "n" resistors of same value are in series, $R_{eq} = nR$

1.5.2. Parallel Connection of Resistors:

Two or more resistors in a circuit are said to be in Parallel when all the resistors are connected to the same nodes and the same voltage is appearing across all these elements.



Consider the circuit in fig(a), where two resistors are connected in parallel and therefore have the same voltage across them. From Ohm's law,

$$v = i_1 R_1 = i_2 R_2 \quad (1)$$

$$i_1 = v/R_1, \quad i_2 = v/R_2 \quad (2)$$

Applying KCL at node a gives the total current i as

$$i = i_1 + i_2 \quad (3)$$

Substituting Eq. (2) into Eq. (3), we get $i = v/R_1 + v/R_2 = v(1/R_1 + 1/R_2) = v/R_{eq}$

$$1/R_{eq} = 1/R_1 + 1/R_2$$

$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$$

Note: The equivalent resistance of two parallel resistors is equal to the product resistances divided by their sum.

If "n" resistors are in parallel, $1/R_{eq} = 1/R_1 + 1/R_2 + \dots + 1/R_n$

If "n" resistors of

It is often more convenient to use conductance rather than resistance when dealing with resistors in parallel. The equivalent conductance for N resistors in parallel is

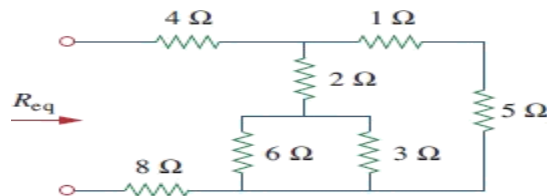
$$G_{eq} = G_1 + G_2 + G_3 + \dots + G_N$$

$$\text{where } G_{eq} = 1/R_{eq}, \quad G_1 = 1/R_1, \quad G_2 = 1/R_2, \quad G_3 = 1/R_3, \dots, \quad G_N = 1/R_N.$$

Note: The equivalent conductance of resistors connected in parallel is the sum of their individual conductance.

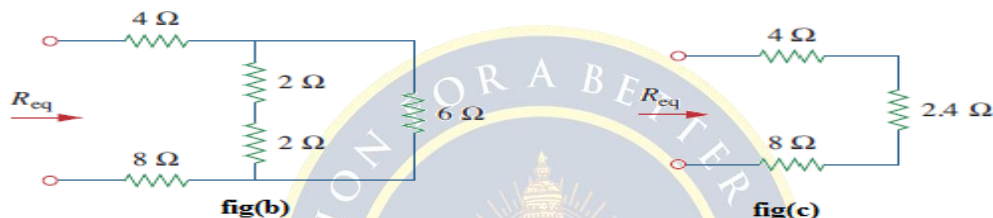
Example Problems:

1) Find the Req for the circuit shown in below figure.



fig(a)

Solution:



To get Req we combine resistors in series and in parallel. The 6 ohms and 3 ohms resistors are in parallel, so their equivalent resistance is

$$6\Omega \parallel 3\Omega = \frac{6 \times 3}{6 + 3} = 2\Omega$$

Also, the 1 ohm and 5ohms resistors are in series; hence their equivalent resistance is

$$1\Omega + 5\Omega = 6\Omega$$

Thus the circuit in Fig.(b) is reduced to that in Fig. (c). In Fig. (b), we notice that the two 2 ohms resistors are in series, so the equivalent resistance is

$$2\Omega + 2\Omega = 4\Omega$$

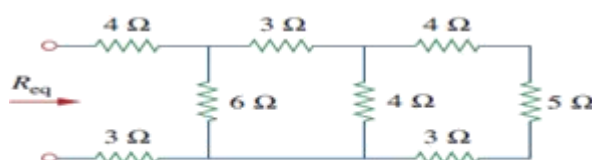
This 4 ohms resistor is now in parallel with the 6 ohms resistor in Fig.(b); their equivalent resistance is

$$4\Omega \parallel 6\Omega = \frac{4 \times 6}{4 + 6} = 2.4\Omega$$

The circuit in Fig.(b) is now replaced with that in Fig.(c). In Fig.(c), the three resistors are in series. Hence, the equivalent resistance for the circuit is

$$R_{eq} = 4\Omega + 2.4\Omega + 8\Omega = 14.4\Omega$$

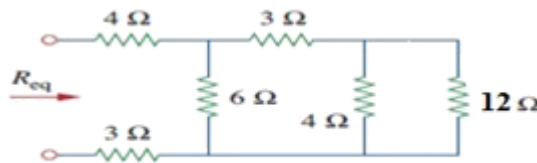
2) Find the Req for the circuit shown in below figure.



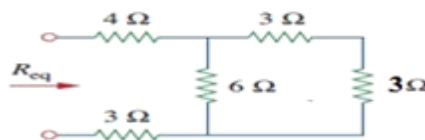
fig(a)

Solution:

In the given network 4 ohms, 5 ohms and 3 ohms comes in series then equivalent resistance is $4+5+3=12$ ohms

**fig(b)**

From fig(b), 4 ohms and 12 ohms are in parallel, equivalent is 3 ohms

**fig(c)**

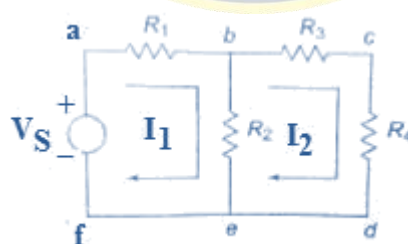
From fig(c), 3 ohms and 3 ohms are in series, equivalent resistance is 6 ohms

**fig(d)**

From fig(d), 6 ohms and 6 ohms are in parallel, equivalent resistance is 3 ohms

**fig(e)****1.6 MESH ANALYSIS with example:**

Determination of mesh currents:

**Solution:**

Step (1): Identify the no. of meshes in the given circuit. There are two meshes.

Mesh(1)**abef**and

Mesh(2)**bcde**

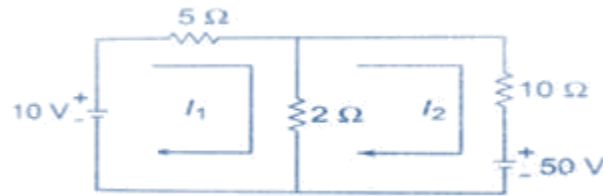
Step (2): Apply the KVL to the all meshes.

For mesh (1) by applying KVL... $V_s - I_1 \cdot R_1 + (I_1 - I_2) \cdot R_2 = 0$(1)

For mesh (2) by applying KVL.... $I_2 \cdot (R_3 + R_4) + (I_2 - I_1) \cdot R_2 = 0$(2)

Step (3): solve the above equations for mesh currents.

Problem: Write down the mesh current equations for the circuit shown in the figure below and determine the currents I_1 and I_2 .



Solution:

By applying KVL to the two meshes,

we get $5 I_1 + 2(I_1 - I_2) = 10$

$10 I_2 + 2(I_2 - I_1) = -50$.

Solving the above equations gives $I_1 = 0.25 \text{ A}$ and $I_2 = -4.25 \text{ A}$. The negative sign for the current I_2 indicates that it flows in the opposite direction to that assumed in the loop two.

Example: Determine the current in the 5Ω resistor shown in the figure below.



Solution:

Step(1): Here the current source exists between mesh(2) and mesh(3). Hence, super mesh is the combination of mesh(2) and mesh(3). Applying KVL to the super mesh (combination of mesh 2 and mesh 3 after removing the branch with the current source of 2 A and resistance of 3Ω) we get :

$$10(I_2 - I_1) + 2I_2 + I_3 + 5(I_3 - I_1) = 0$$

$$-15I_1 + 12I_2 + 6I_3 = 0 \dots\dots\dots (1)$$

Step (2): Applying KVL first to the normal mesh 1 we get :

$$10(I_1 - I_2) + 5(I_1 - I_3) = 50$$

$$15I_1 - 10I_2 - 5I_3 = 50 \dots\dots\dots (2)$$

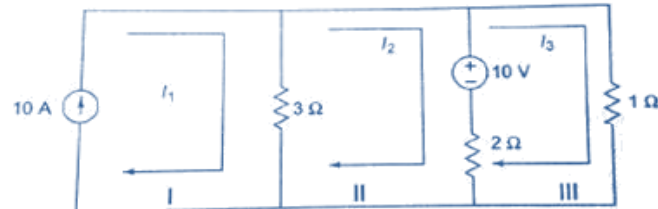
Step (3): We can get the third equation from the relation between the current source of 2 A , and currents I_2 & I_3 as :

$$I_2 - I_3 = 2A \dots\dots\dots (3)$$

Step (4): Solving the above three equations for I_1 , I_2 and I_3 we get $I_1 = 19.99$ A $I_2 = 17.33$ A and $I_3 = 15.33$ A

The current in the $5\ \Omega$ resistance = $I_1 - I_3 = 19.99 - 15.33 = 4.66$ A

Example: Write down the mesh equations for the circuit shown in the figure below and find out the values of the currents I_1 , I_2 and I_3



Solution: In this circuit the current source is in the perimeter of the circuit and hence the first mesh is ignored. So, here no need to create the super mesh.

Applying KVL to mesh 1 we get :

$$3(I_2 - I_1) + 2(I_2 - I_3) = -10$$

$$-3.I_1 + 5.I_2 - 2.I_3 = -10 \dots\dots\dots (1)$$

Next applying KVL to mesh 2 we get :

$$I_3 + 2(I_3 - I_2) = 10$$

$$-2.I_2 + 3.I_3 = 10 \dots\dots\dots (2)$$

$$\text{And from the first mesh we observe that } I_1 = 10 \text{ A } \dots\dots\dots (3)$$

And solving these three equations we get : $I_1 = 10$ A, $I_2 = 7.27$ A, $I_3 = 8.18$ A

1.7 NODAL ANALYSIS:

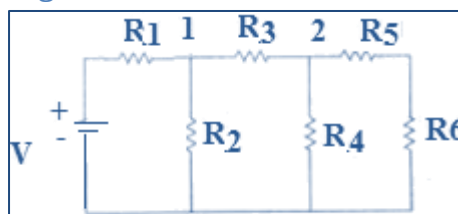
Nodal analysis provides another general procedure for analyzing circuits nodal voltages as the circuit variables. It is preferably useful for the circuits that have many no. of nodes. It is applicable for the both planar and non planar circuits. This analysis is done by using KCL and Ohm's law.

Node: It is a junction at which two or more branches are interconnected.

Simple Node: Node at which only two branches are interconnected.

Principal Node: Node at which more than two branches are interconnected. **Nodal analysis with example:**

Determination of node voltages:

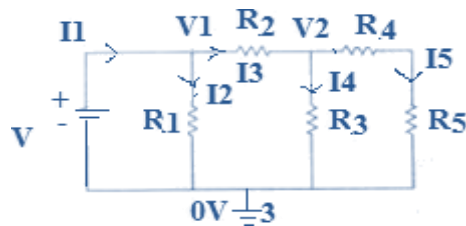


Procedure:

Step (1): Identify the no. nodes, simple nodes and principal nodes in the given circuit. Among all the nodes one node is taken as reference node. Generally bottom is taken as reference node. The potential at the reference node is 0v.

In the given circuit there are 3 principal nodes in which node (3) is the reference node.

Step (2): Assign node voltages to the all the principal nodes except reference node and assign branch currents to all branches.



Step (3): Apply KCL to those principal nodes for nodal equations and by using ohm's law express the node voltages in terms of branch current.

Applying KCL to node(1)----- $1 = I_2 + I_3$

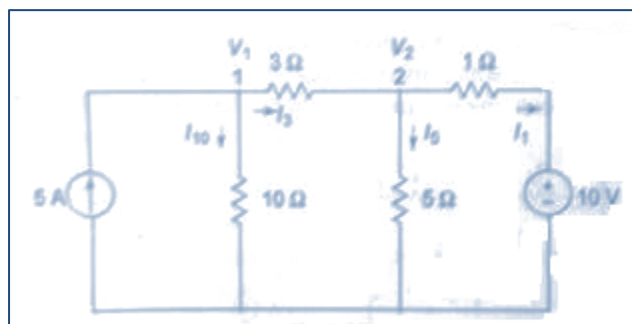
Using ohm's law, we get $(V - V_1)/R_1 = (V_2 - 0)/R_2 + (V_1 - V_2)/R_3$ (1)

Applying KCL to node(2)----- $3 = I_4 + I_5$

Using ohm's law, we get $(V_1 - V_2)/R_3 = (V_2 - 0)/R_4 + (V_2 - 0)/R_5$ (2)

Step(4): Solve the above nodal equations to get the node voltages.

Example: Write the node voltage equations and find out the currents in each branch of the circuit shown in the figure below.

**Solution:**

The node voltages and the directions of the branch currents are assigned as shown in given figure.

Applying KCL to node 1, we get: $5 = I_{10} + I_3$

$$5 = (V_1 - 0)/10 + (V_1 - V_2)/3$$

$$V1(13/30) - V2(1/3) = 5 \dots\dots\dots (1)$$

Applying KCL to node 2, we get: $I_3 = I_5 + I_1$

$$(V1 - V2)/3 = (V2 - 0)/5 + (V2 - 10)/1$$

$$V1(1/3) - V2(23/15) = -10 \dots\dots\dots (2)$$

Solving these two equations for $V1$ and $V2$ we get :

$V1 = 19.85 \text{ V}$ and $V2 = 10.9 \text{ V}$ and the currents are :

$$I_{10} = V1/10 = 1.985 \text{ A}$$

$$I_3 = (V1 - V2)/3 = (19.85 - 10.9)/3 = 2.98 \text{ A} \quad I_5 = V2/5 = 10.9/5 = 2.18 \text{ A}$$

$$I_1 = (V2 - 10) = (10.9 - 10)/1 = 0.9 \text{ A}$$

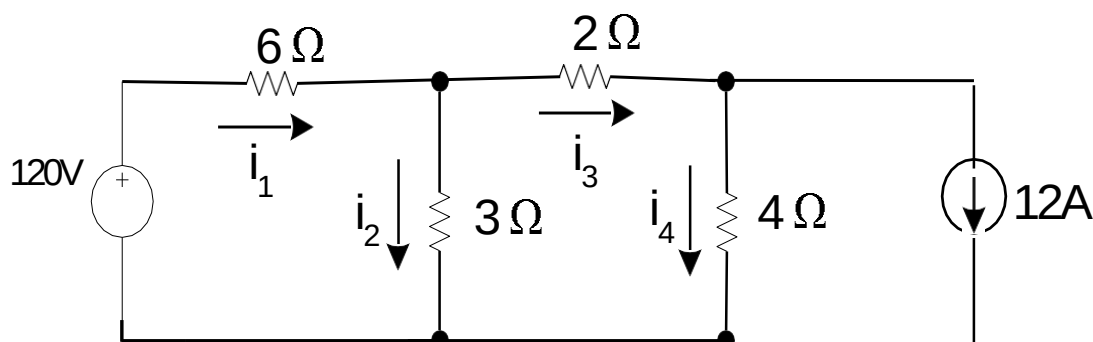
1.8 SUPERPOSITION THEOREM:

- It is used to find the solution to networks with two or more sources that are not in series or parallel
- The current through, or voltage across, an element in a linear bilateral network is equal to the algebraic sum of the currents or voltages produced independently by each source.
- For a two-source network, if the current produced by one source is in one direction, while that produced by the other is in the opposite direction through the same resistor, the resulting current is the difference of the two and has the direction of the larger
- If the individual currents are in the same direction, the resulting current is the sum of two in the direction of either current.
- The total power delivered to a resistive element must be determined using the total current through or the total voltage across the element and cannot be determined by a simple sum of the power levels established by each source

For applying Superposition theorem:-

- Replace all other independent voltage sources with a short circuit (thereby eliminating difference of potential. i.e. $V=0$, internal impedance of ideal voltage source is ZERO (short circuit)).
- Replace all other independent current sources with an open circuit (thereby eliminating current. i.e. $I=0$, internal impedance of ideal current source is infinite (open circuit)).

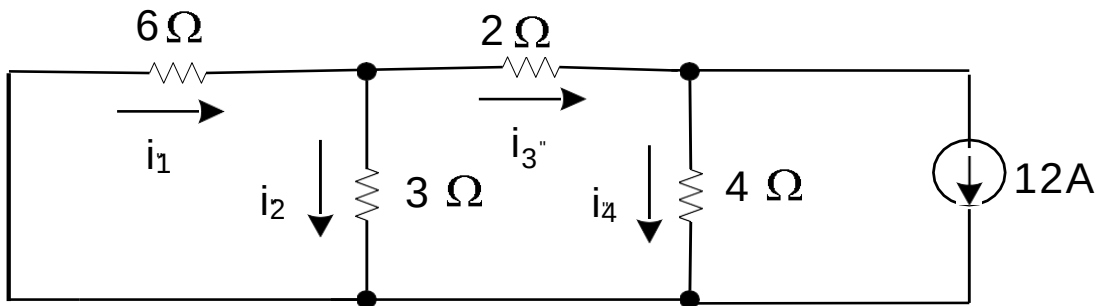
Example:- Determine the branches current using Superposition theorem.



Solution:

Figure1

The application of the superposition theorem is shown in Figure 1, where it is used to calculate the branch current. We begin by calculating the branch current caused by the voltage source of 120 V. By substituting the ideal current with open circuit, we deactivate the current source, as shown in Figure 2.



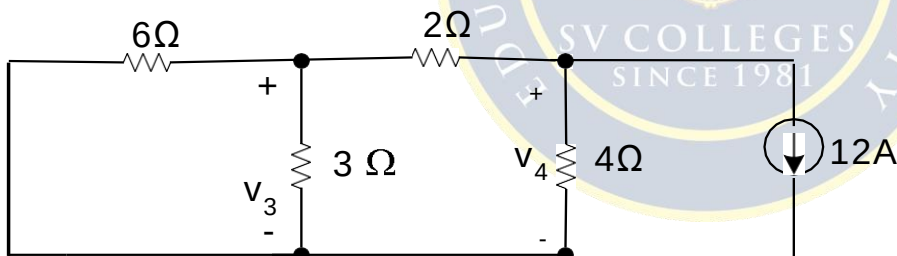
$$i'_1 = \frac{120-30}{6} = 15\text{A}$$

$$i'_2 = \frac{30}{3} = 10\text{A}$$

$$i'_3 = i'_4 = \frac{30}{6} = 5\text{A}$$

In order to calculate the current cause by the current source, we deactivate the ideal voltage source with a short circuit, as shown

To determine the branch current, solve the node voltages across the 3Ω dan 4Ω resistors as shown in Figure



$$\frac{v_3}{3} + \frac{v_3}{6} + \frac{v_3-v_4}{2} = 0$$

$$\frac{v_4-v_3}{2} + \frac{v_4}{6} + 12 = 0$$

The two voltages are

By solving these equations, we obtain

$$v_3 = -12\text{ V } v_4 = -24\text{ V}$$

Now we can find the branches current,

$$i_1'' = \frac{-v_3}{6} = \frac{12}{6} = 2A$$

$$i_2'' = \frac{v_3}{3} = \frac{-12}{3} = -4A$$

$$i_3'' = \frac{v_3 - v_4}{2} = \frac{-12 + 24}{2} = 6A$$

$$i_4'' = \frac{v_4}{4} = \frac{-24}{4} = -6A$$

1.9 AC CIRCUITS

Introduction (Different Periodic Wave Forms)

Almost 90% of the electrical energy we use in our day-to-day life is from alternating source. Be it our home appliances, office equipment or industrial machines, we use AC source to power these devices.

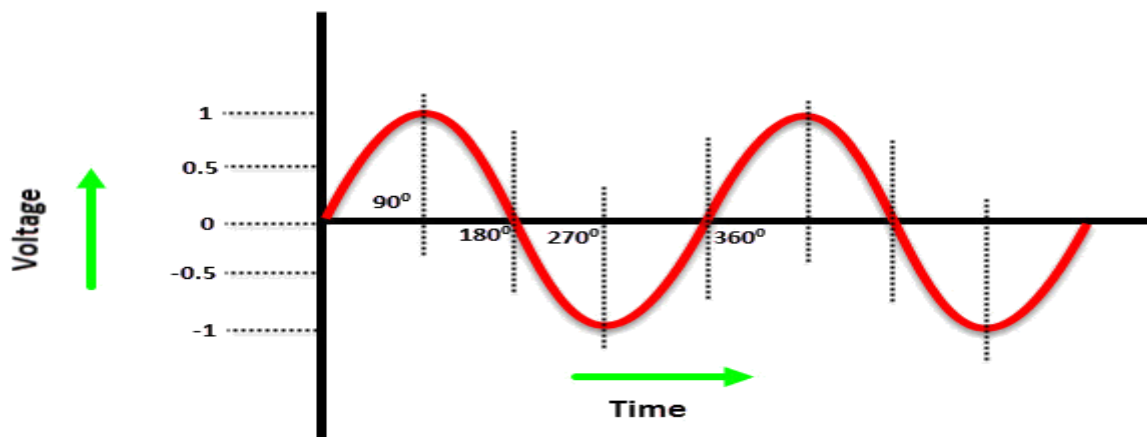
If you are a beginner, then Alternating Current or simply AC is a type of electric power in which the electric current changes periodically, both magnitude and direction. Further, depending on the application, AC Power can be delivered in either a Single Phase or a Three Phase system.

A Single Phase AC Power system consists of two wires known as the phase (or sometimes a Line or Live) and the neutral wire. In case a three phase system, you use either three wires or four wires for transmitting power (no neutral in three wire three phase power).

Let us now go into the details of single phase and three phase systems and also see the difference between single phase and three phase power supplies.

What is Single Phase Power Supply?

As mentioned earlier, in a Single Phase Power Supply, the power is distributed using only two wires called Phase and neutral. Since AC Power takes the shape of a sinusoidal wave, the voltage in a single phase supply peaks at 900 during the positive cycle and again at 2700 during the negative cycle.



The phase wire carries the current to the load and the neutral wire provides the return path of the current. Usually, the single phase voltage is 230V and the frequency is 50Hz (depending on where you live).

Since the voltage in a single phase supply rises and falls, a constant power cannot be delivered to the load.

Advantages

- It is very common form of power supply to most small power requirement. Almost all residential supplies are single phase supplies as the domestic appliances require a small amount of power to run lights, fans, coolers, heaters, small air conditioners etc.
- The design and operation of a single phase power supply system is often simple.
- Depending on the region, a single phase supply is sufficient for loads up to 2500 Watts.

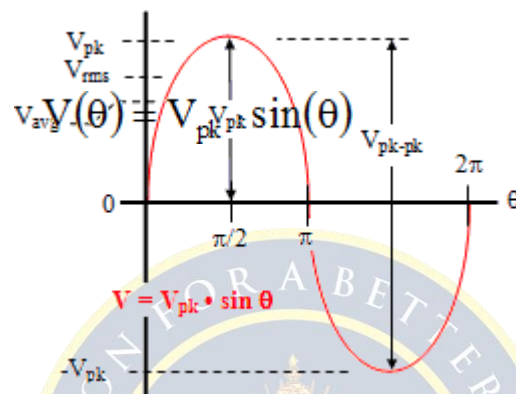
Disadvantages

- Small single phase motors (usually less than 1kW) cannot start directly with the help of a single phase supply as there isn't sufficient initial torque for the motor. So, additional circuitry like a Motor Starters (like a starter capacitor in fans and pumps) are needed for proper operation.
- Heavy loads like industrial motors and other equipment cannot be run on a single phase supply.
- Average value, RMS Value, Form factor and Peak factor for different waveforms:

Sinusoidal wave:

- A sinewave is defined by the trigonometric sine function. When plotted as voltage (V) as a function of phase (θ), it looks similar to the figure to the below. The waveform repeats every 2π radians (360°), and is symmetrical about the voltage axis (when no DC offset is present). Voltage and current exhibiting cyclic behavior is referred to as alternating; i.e., alternating current (AC). One full cycle is shown here. The basic equation for a sinewave is as follows:

- There are a number of ways in which the amplitude of a sinewave is referenced, usually as peak voltage (V_{pk} or V_p), peak-to-peak voltage (V_{pp} or V_{p-p} or V_{pkpk} or V_{pk-pk}), average voltage (V_{av} or V_{avg}), and root-mean-square voltage (V_{rms}). Peak voltage and peak-to-peak voltage are apparent by looking at the above plot. Root-mean-square and average voltage are not so apparent.



Average Voltage (V_{avg})

- As the name implies, V_{avg} is calculated by taking the average of the voltage in an appropriately chosen interval. In the case of symmetrical waveforms like the sinewave, a quarter cycle faithfully represents all four quarter cycles of the waveform. Therefore, it is acceptable to choose the first quarter cycle, which goes from 0 radians (0°) through $\pi/2$ radians (90°).
- As with the V_{rms} formula, a full derivation for the V_{avg} formula is given here as well.

$$\begin{aligned}
 V_{avg} &= \frac{1}{\pi/2} \cdot \int_0^{\pi/2} V_{pk} \cdot \sin \theta \cdot d\theta = \frac{2}{\pi} \cdot V_{pk} \cdot -\cos \theta \Big|_0^{\pi/2} \\
 &= \frac{-2}{\pi} \cdot V_{pk} \cdot \left(\cos \frac{\pi}{2} - \cos 0 \right) = \frac{-2}{\pi} \cdot V_{pk} \cdot (0 - 1) = \frac{2}{\pi} \cdot V_{pk} \\
 \text{So, } V_{avg} &= \frac{2}{\pi} \cdot V_{pk} \approx 0.636 V_{pk},
 \end{aligned}$$

Root-Mean-Square Voltage (Vrms)

As the name implies, Vrms is calculated by taking the square root of the mean average of the square of the voltage in an appropriately chosen interval. In the case of symmetrical waveforms like the sine wave, a quarter cycle faithfully represents all four quarter cycles of the waveform. Therefore, it is acceptable to choose the first quarter cycle, which goes from 0 radians (0°) through $\pi/2$ radians (90°).

Vrms is the value indicated by the vast majority of AC voltmeters. It is the value that, when applied across a resistance, produces that same amount of heat that a direct current (DC) voltage of the same magnitude would produce. For example, 1 V applied across a 1 Ω resistor produces 1 W of heat. A 1 Vrms sine wave applied across a 1 Ω resistor also produces 1 W of heat. That 1 Vrms sine wave has a peak voltage of $\sqrt{2}$ V (≈ 1.414 V), and a peak-to-peak voltage of $2\sqrt{2}$ V (≈ 2.828 V). Since finding a full derivation of the formulas for root-mean-square (Vrms) voltage is difficult, it is done here for you.

$$\begin{aligned}
 V_{\text{rms (sinewave)}} &= \sqrt{\frac{1}{\pi/2} \int_0^{\pi/2} (V_{\text{pk}} \sin \theta)^2 \cdot d\theta} = \sqrt{\frac{2 \cdot V_{\text{pk}}^2}{\pi} \left(\frac{\theta}{2} - \frac{1}{4} \sin(2 \cdot \theta) \right) \bigg|_0^{\pi/2}} \\
 &= \frac{\sqrt{2} \cdot V_{\text{pk}}}{\sqrt{\pi}} \sqrt{\left(\frac{\pi/2}{2} - \frac{1}{4} \sin(\pi) \right) - \left(\frac{0}{2} - \frac{1}{4} \sin(0) \right)} \\
 &= \frac{\sqrt{2} \cdot V_{\text{pk}}}{\sqrt{\pi}} \sqrt{\left(\frac{\pi}{4} - 0 \right) - (0 - 0)} = \frac{\sqrt{2} \cdot V_{\text{pk}}}{\sqrt{\pi}} \cdot \frac{\sqrt{\pi}}{2} = \frac{1}{\sqrt{2}} \cdot V_{\text{pk}}
 \end{aligned}$$

$$\text{So, } V_{\text{rms}} = \frac{1}{\sqrt{2}} \cdot V_{\text{pk}} \approx 0.707 V_{\text{pk}}$$

Form factor:

- Two alternating periodic waveforms of the same amplitude and frequency may look different depending upon their wave shape/form and then their average & RMS values will be different. In order to compare such different waveforms of the same frequency and amplitude but of different wave shape a parameter called Form factor is defined as the ratio of it's RMS and Average values.

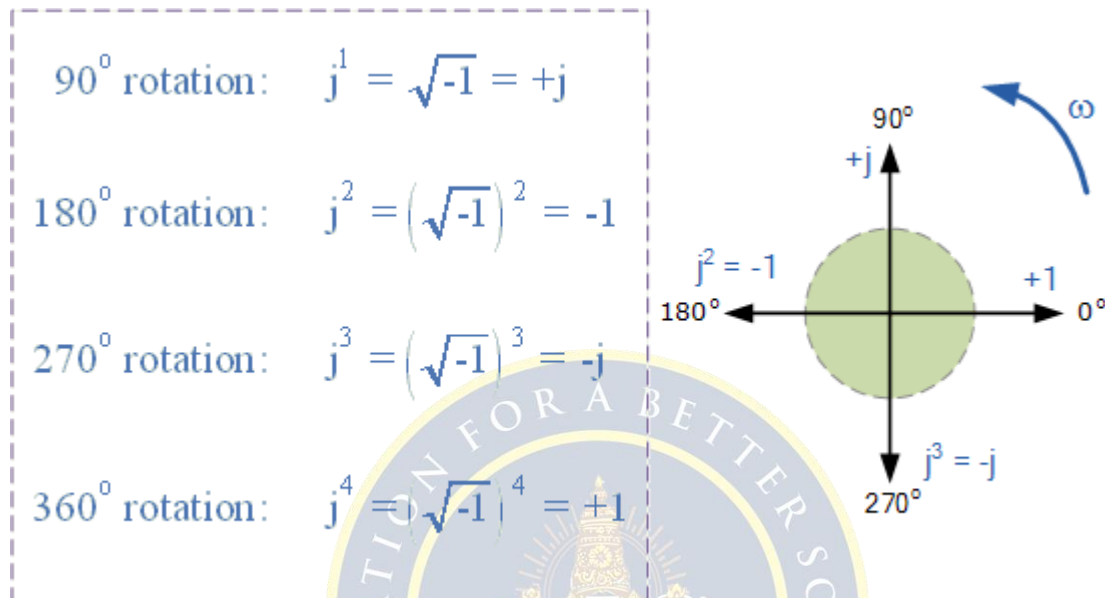
For a sinusoidal signal of peak voltage Vm it is given by :

Form factor of a sinusoidal signal = $V_{\text{rms}} / V_{\text{av}} = 0.707 V_m / 0.637 V_m = 1.11$

Peak Factor (Or Crest factor): Is defined as the ratio of maximum value to the R.M.S value of an alternating quantity.

$$\begin{aligned}\text{Peak factor of a sinusoidal signal} &= V_{\max}/V_{\text{rms}} \\ &= V_{\max}/(0.707 V_m) \\ &= 1.414\end{aligned}$$

Vector Rotation



So by multiplying an imaginary number by j^2 will rotate the vector by 180° anticlockwise, multiplying by j^3 rotates it 270° and by j^4 rotates it 360° or back to its original position. Multiplication by j^{10} or by j^{30} will cause the vector to rotate anticlockwise by the appropriate amount. In each successive rotation, the magnitude of the vector always remains the same.

Complex and Polar forms of Representation:

In Electrical Engineering there are different ways to represent a complex number either graphically or mathematically. One such way that uses the cosine and sine rule is called the Cartesian or Rectangular Form.

A complex number is represented by a real part and an imaginary part that takes the generalised form of:

$$Z = x + jy$$

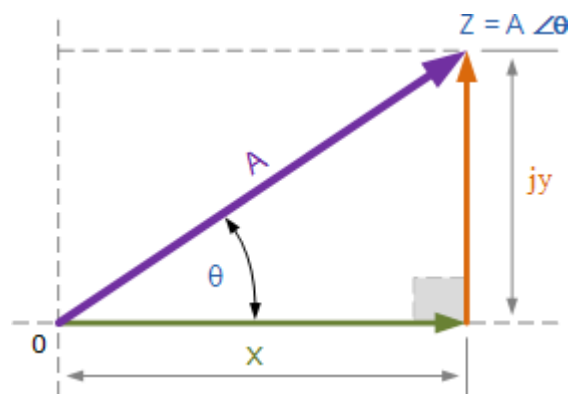
Where

Z - is the Complex Number representing the Vector x - is the Real part or the Active component

y - is the Imaginary part or the Reactive component j - is defined by $\sqrt{-1}$

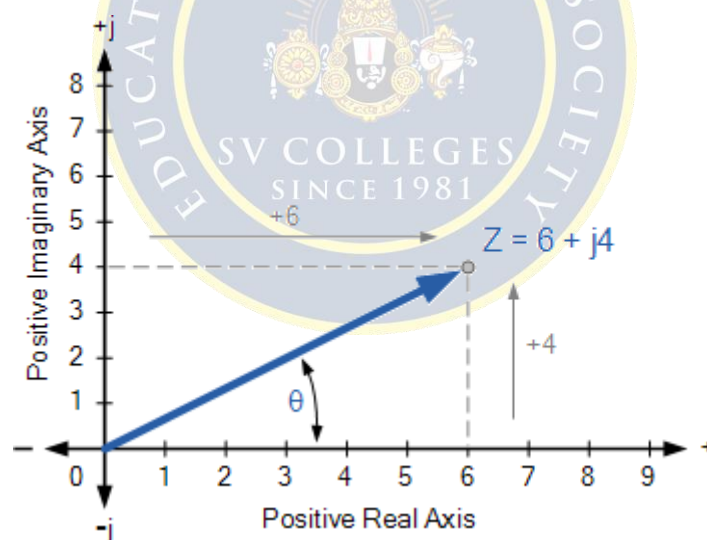
In the rectangular form, a complex number can be represented as a point on a two-dimensional plane called the complex or s-plane. So for example, $Z = 6 + j4$ represents a single point whose coordinates represent 6 on the horizontal real axis and 4 on the vertical imaginary axis as shown.

Complex Numbers using the Complex or s-plane:



Complex Numbers using Polar Form:

Unlike rectangular form which plots points in the complex plane, the Polar Form of a complex number is written in terms of its magnitude and angle. Thus, a polar form vector is presented



as: $Z = A \angle \pm \theta$, where: Z is the complex number in polar form, A is the magnitude or modulo of the vector and θ is its angle or argument of A which can be either positive or negative. The magnitude and angle of the point still remains the same as for the rectangular form above, this time in polar form the location of the point is represented in a “triangular form” as shown below.

Polar Form Representation of a Complex Number:

As the polar representation of a point is based around the triangular form, we can use simple geometry of the triangle and especially trigonometry and Pythagoras's Theorem on triangles to find both the magnitude and the angle of the complex number. As we remember from school, trigonometry deals with the relationship between the sides and the angles of triangles so we can describe the relationships between the sides as:

$$A^2 = X^2 + Y^2 \quad A = \sqrt{X^2 + Y^2}$$

$$\text{Also } X = A \cos \theta \quad Y = A \sin \theta$$

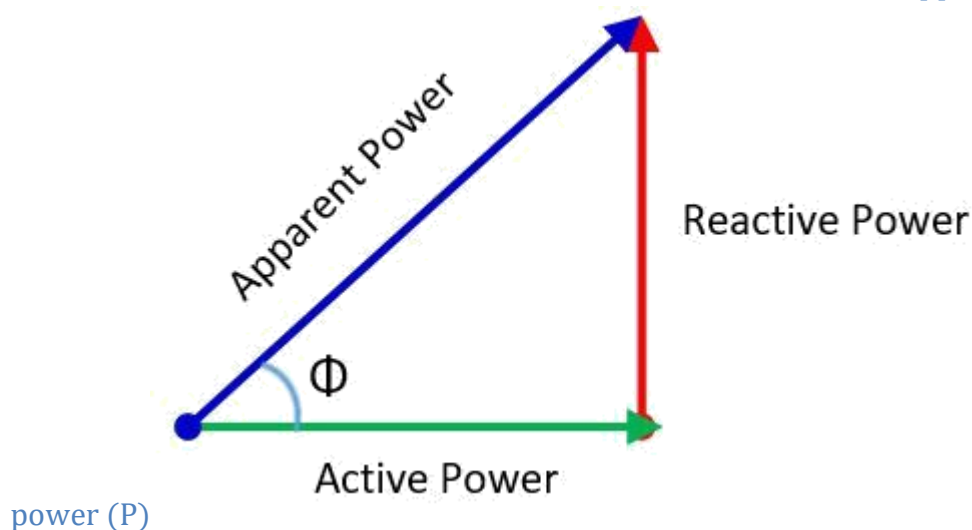
Using trigonometry again, the angle θ of A is given as follows.

$$\theta = \tan^{-1} y/x$$

Then in Polar form the length of A and its angle represents the complex number instead of a point. Also in polar form, the conjugate of the complex number has the same magnitude or modulus it is the sign of the angle that changes, so for example the conjugate of $6 \angle 30^\circ$ would be $6 \angle -30^\circ$.

1.10 ACTIVE, REACTIVE & APPARENT POWER

Difference between Active & Reactive Power: The most significant difference between the active and reactive power is that the active power is the actual power which is dissipated in the circuit. Whereas, the reactive power is the useless power which only flows between the source and load. The other differences between the active and reactive power are explained below in the comparison chart. The active, apparent and real power induces in the circuit only when their current lags behind the applied voltage by an angle of Φ . The right-angled triangle shown below shows the relation between the active, reactive and apparent power.



power-factor

Where, S – apparent power

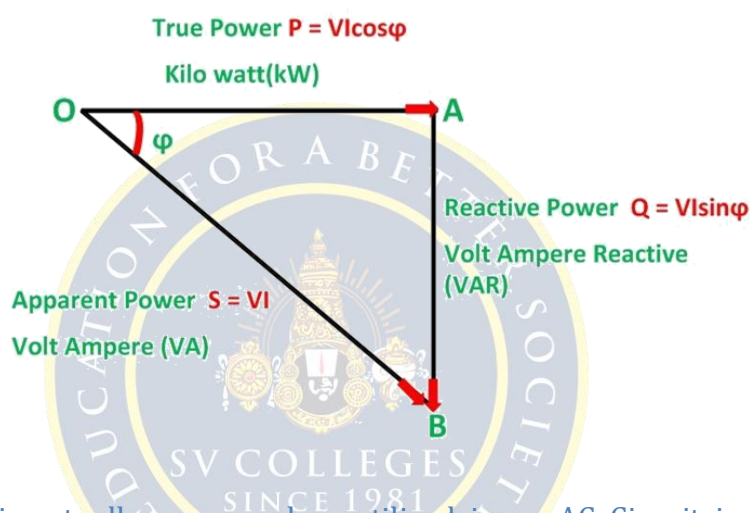
Q – reactive power

P – Active power

Power Triangle

Power Triangle is the representation of a right angle triangle showing the relation between active power, reactive power and apparent power.

When each component of the current that is the active component ($I\cos\phi$) or the reactive component ($I\sin\phi$) is multiplied by the voltage V, a power triangle is obtained shown in the figure below:



The power which is actually consumed or utilized in an AC Circuit is called True power or Active Power or real power. It is measured in kilowatt (kW) or MW.

The power which flows back and forth that means it moves in both the direction in the circuit or reacts upon it, is called Reactive Power. The reactive power is measured in kilovolt-ampere reactive (kVAR) or MVAR.

The product of root mean square (RMS) value of voltage and current is known as Apparent Power. This power is measured in KVA or MVA.

The following point shows the relationship between the following quantities and is explained by graphical representation called Power Triangle shown above.

When an active component of current is multiplied by the circuit voltage V, it results in active power. It is this power which produces torque in the motor, heat in the heater, etc. This power is measured by the wattmeter.

When the reactive component of the current is multiplied by the circuit voltage, it gives reactive power. This power determines the power factor, and it flows back and forth in the circuit.

When the circuit current is multiplied by the circuit voltage, it results in apparent power.

From the power triangle shown above the power, the factor may be determined by taking the ratio of true power to the apparent power.

$$\text{Power factor } \cos\phi = \frac{\text{Active Power}}{\text{Apparent Power}} = \frac{KW}{KVA}$$

As we know simply power means the product of voltage and current but in AC circuit except for pure resistive circuit there is usually a phase difference between voltage and current and thus VI does not give real or true power in the circuit.

Definition of Active Power:

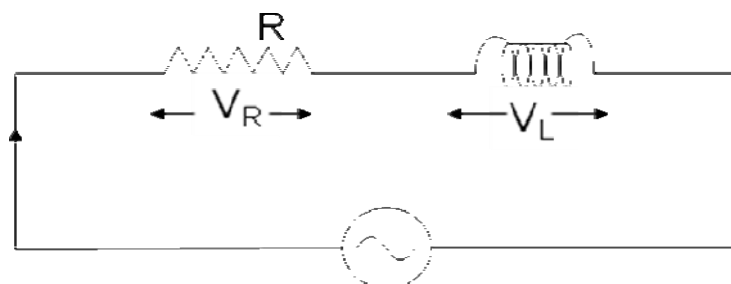
The power which is dissipated or do the useful work in the circuit is known as the active power. It is measured in watts or megawatts. The active power is denoted by the capital alphabet P. The average value of power in the circuit is given by the expression.

Definition of Reactive Power:

The reactive power moves between the source and load of the circuit. This power is not doing any useful works on the load. Q represents the reactive power, and it is measured in VAR. The reactive power is stored in the circuit, and it is discharged by the induction motor, transformer or by solenoids.

1.11 ANALYSIS OF SINGLE-PHASE AC CIRCUITS CONSISTING OF RL - RC - RLC SERIES CIRCUITS.

R-L Series circuit



Consider an AC circuit with a resistance R and an inductance L connected in series as shown in the figure. The alternating voltage v is given by

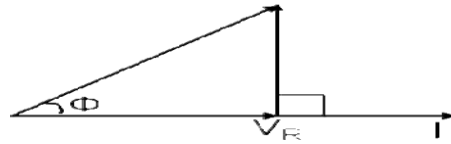
$$v = V_m \sin \omega t$$

The current flowing in the circuit is i . The voltage across the resistor is V_R and that across the inductor is V_L .

$V_R = IR$ is in phase with I

$V_L = IX_L$ leads current by 90 degrees

With the above information, the phasor diagram can be drawn as shown



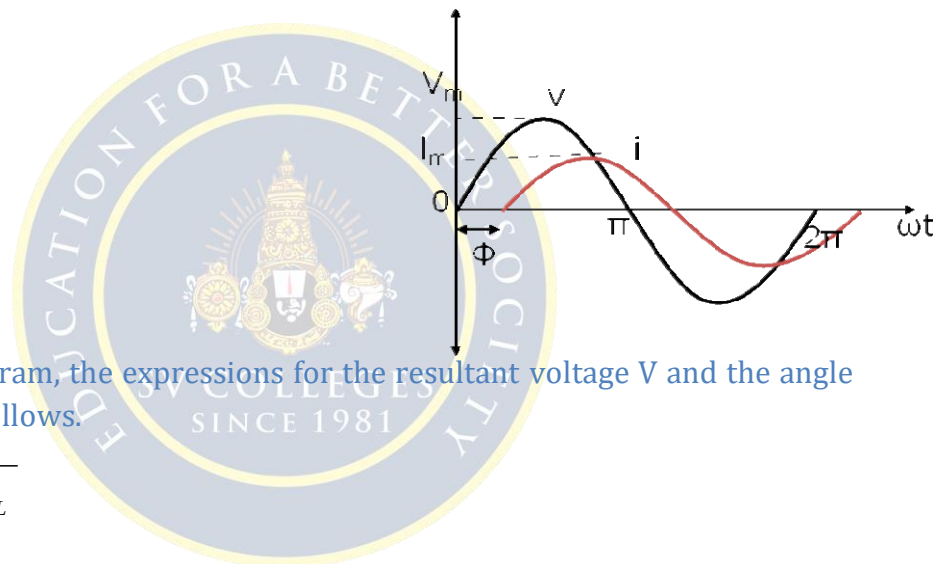
The current I is taken as the reference phasor. The voltage V_R is in phase with I and the voltage V_L leads the current by 90° . The resultant voltage V can be drawn as shown in the

figure. From the phasor diagram we observe that the voltage leads the current by an angle Φ or in other words the current lags behind the voltage by an angle Φ .

The waveform and equations for an RL series circuit can be drawn as below.

$$V = V_m \sin \omega t$$

$$I = I_m \sin(\omega t - \Phi)$$



From the phasor diagram, the expressions for the resultant voltage V and the angle Φ can be derived as follows.

$$V = \sqrt{V_R^2 + V_L^2}$$

$$V_R = IR, V_L = IX_L$$

$$V = \sqrt{(IR)^2 + (IX_L)^2}$$

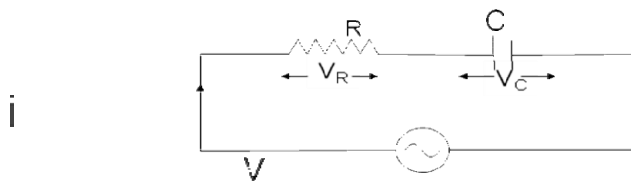
$$V = I \sqrt{R^2 + X_L^2}$$

$$V = IZ$$

Where impedance

$$Z = \sqrt{R^2 + X_L^2}$$

R-C Series circuit



Consider an AC circuit with a resistance R and a capacitance C connected in series as shown in the figure. The alternating voltage v is given by

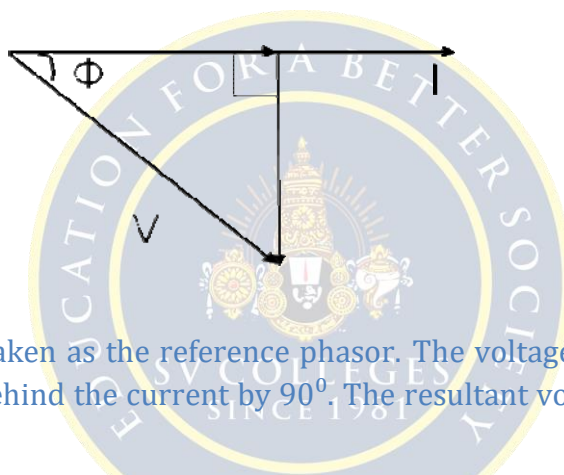
$$V = V_m \sin \omega t$$

The current flowing in the circuit is i . The voltage across the resistor is V_R and that across the capacitor is V_C .

$V_R = IR$ is in phase with I

$V_C = IX_C$ lags behind the current by 90°

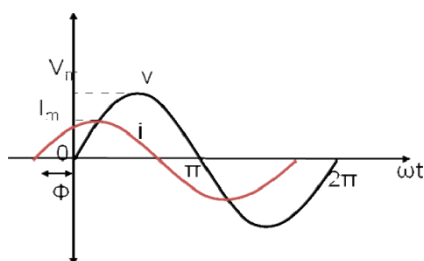
With the above information, the phasor diagram can be drawn as



The current I is taken as the reference phasor. The voltage V_R is in phase with I and the voltage V_C lags behind the current by 90° . The resultant voltage V can be drawn as shown in

the figure. From the phasor diagram we observe that the voltage lags behind the current by an angle Φ or in other words the current leads the voltage by an angle Φ .

The waveform and equations for an RC series circuit can be drawn as



$$V = V_m \sin \omega t$$

$$I = I_m \sin(\omega t + \Phi)$$

From the phasor diagram, the expressions for the resultant voltage V and the angle Φ can be derived as follows.

$$V = \sqrt{V_R^2 + V_C^2}$$

$$V_R = IR$$

$$V_C = I X_C$$

$$V = \sqrt{(IR)^2 + (IX_C)^2}$$

$$V = I \sqrt{R^2 + X_C^2}$$

$$V = IZ$$

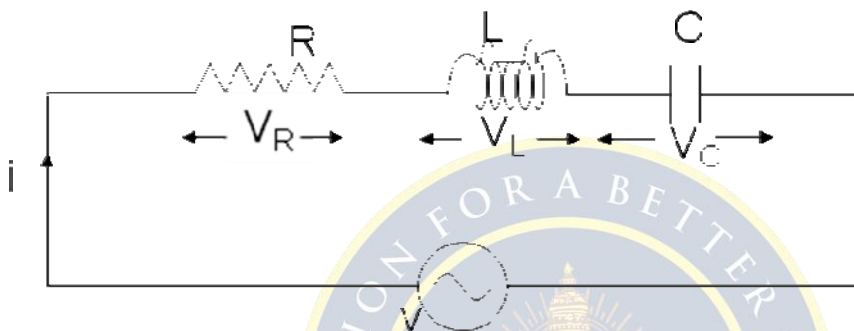
$$\text{Where impedance } Z = \sqrt{R^2 + X_C^2}$$

Average power

$$P = VI \cos \Phi$$

Hence the power in an RC series circuit is consumed only in the resistance. The capacitance does not consume any power

R-L-C Series circuit



Consider an AC circuit with a resistance R, an inductance L and a capacitance C connected in series as shown in the figure. The alternating voltage v is given

$$V = V_m \sin \omega t$$

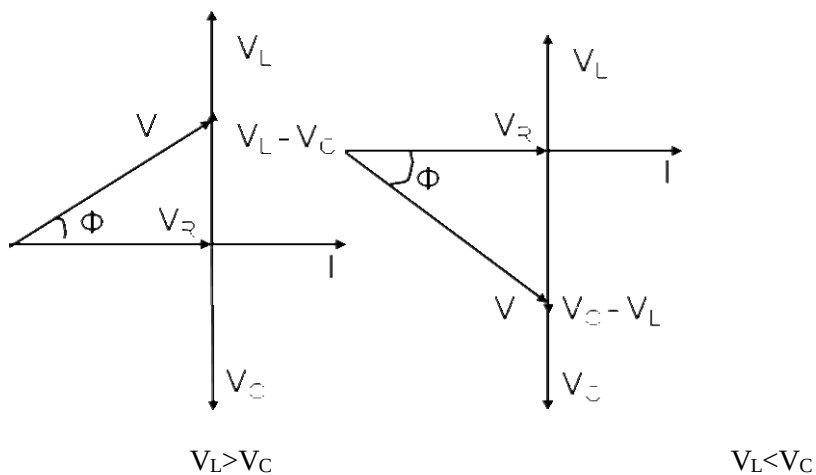
The current flowing in the circuit is i . The voltage across the resistor is V_R , the voltage across the inductor is V_L and that across the capacitor is V_C .

$V_R = IR$ is in phase with I

$V_L = I X_L$ leads the current by 90 degrees

$V_C = I X_C$ lags behind the current by 90 degrees

With the above information, the phasor diagram can be drawn as shown. The current I is taken as the reference phasor. The voltage V_R is in phase with I , the voltage V_L leads the current by 90° and the voltage V_C lags behind the current by 90° . There are two cases that can occur $V_L > V_C$ and $V_L < V_C$ depending on the values of X_L and X_C . And hence there are two possible phasor diagrams. The phasor $V_L - V_C$ or $V_C - V_L$ is drawn and then the resultant voltage V is drawn.



From the phasor diagram we observe that when $V_L > V_C$, the voltage leads the current by an angle Φ or in other words the current lags behind the voltage by an angle Φ . When $V_L < V_C$, the voltage lags behind the current by an angle Φ or in other words the current leads the voltage by an angle Φ .

From the phasor diagram, the expressions for the resultant voltage V and the angle Φ can be derived as follows

$$V = \sqrt{V_R^2 + (V_L - V_C)^2}$$

$$V = \sqrt{(IR)^2 + (IX_L - IX_C)^2}$$

Where impedance $Z = \sqrt{R^2 + (X_L - X_C)^2}$

From the expression for phase angle, we can derive the following three cases

Case (i): When $X_L > X_C$

The phase angle Φ is positive and the circuit is inductive. The circuit behaves like a series RL circuit.

Case (ii): When $X_L < X_C$

The phase angle Φ is negative and the circuit is capacitive. The circuit behaves like a series RC circuit.

Case (iii): When $X_L = X_C$

The phase angle $\Phi = 0$ and the circuit is purely resistive. The circuit behaves like a pure resistive circuit.

The voltage and the current can be represented by the following equations. The angle Φ is positive or negative depending on the circuit elements.

$$V = V_m \sin \omega t$$

$$I = I_m \sin(\omega t \pm \Phi)$$

Average Power

$$P = VI \cos\Phi$$

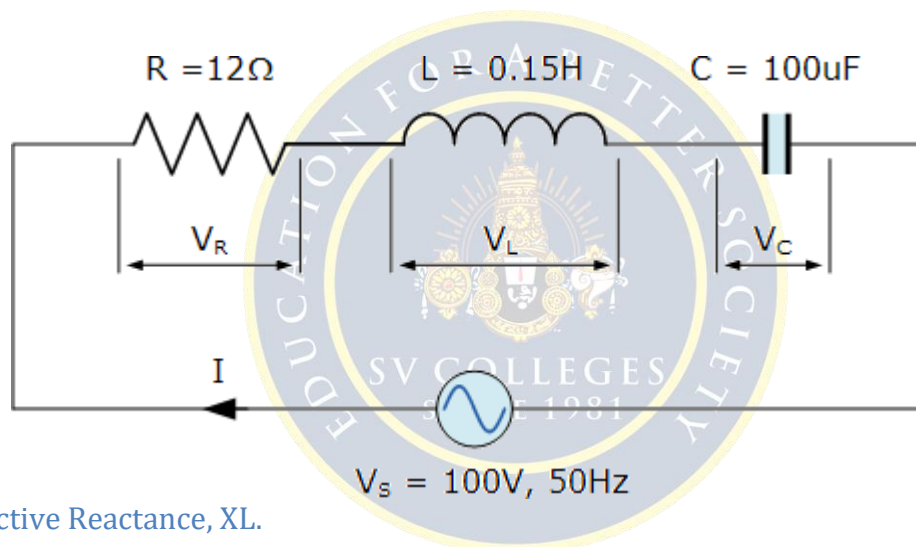
$$P = (IZ) \times I \times \frac{R}{Z}$$

$$P = I^2 R$$

Hence the power in an RLC series circuit is consumed only in the resistance. The inductance and capacitance do not consume any power.

Example:

A series RLC circuit containing a resistance of 12Ω , an inductance of $0.15H$ and a capacitor of $100\mu F$ are connected in series across a $100V$, $50Hz$ supply. Calculate the total circuit impedance, the circuits current, power factor and draw the voltage phasor diagram.



Inductive Reactance, X_L .

$$X_L = 2\pi fL = 2\pi \times 50 \times 0.15 = 47.13\Omega$$

Capacitive Reactance, X_C .

$$X_C = \frac{1}{2\pi fC} = \frac{1}{2\pi \times 50 \times 100 \times 10^{-6}} = 31.83\Omega$$

Circuit Impedance, Z

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

$$Z = \sqrt{12^2 + (47.13 - 31.83)^2}$$

$$Z = \sqrt{144 + 234} = 19.4\Omega$$

Circuits Current, I .

$$I = \frac{V_s}{Z} = \frac{100}{19.4} = 5.14 \text{ Amps}$$

Voltages across the Series RLC Circuit, V_R , V_L , V_C .

$$V_R = I \times R = 5.14 \times 12 = 61.7 \text{ volts}$$

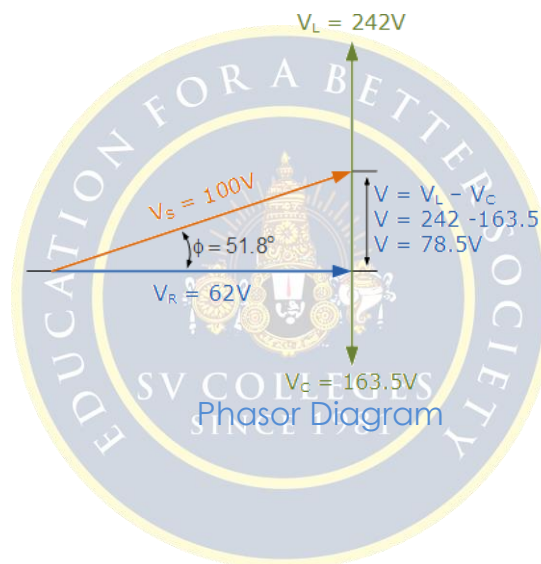
$$V_L = I \times X_L = 5.14 \times 47.13 = 242.2 \text{ volts}$$

$$V_C = I \times X_C = 5.14 \times 31.8 = 163.5 \text{ volts}$$

Circuits Power factor and Phase Angle, θ .

$$\cos \phi = \frac{R}{Z} = \frac{12}{19.4} = 0.619$$

$$\therefore \cos^{-1} 0.619 = 51.8^\circ \text{ lagging}$$



9. Practice Quiz

1. State Ohm's law.

Ohm's law states that the voltage (v) across a resistor is directly proportional to the current (i) flowing through the resistor, at constant temperature. ie, $v \propto i$, $v = iR$, where R is the resistance (Ω).

2. State Kirchoff's Current law.

KCL (Kirchoff's Current Law) states that the algebraic sum of currents entering a node (or a closed boundary) is zero. (or) The sum of the currents entering a node is equal to the sum of the currents leaving the node.

3. State Kirchoff's Voltage law.

KVL (Kirchoff's Voltage Law) states that the algebraic sum of all voltages around a closed path (or loop) is zero. (or) Sum of voltage drop = Sum of voltage rise.

4. An Electric iron is rated 1000W, 240V. Find the current drawn & resistance of the heating element.

$$P = V^2/R ; R = 240^2/1000 = 57.6\Omega \text{ and } I = V/R = 240/57.6 = 4.166 \text{ A}$$

5. Define i) charge ii) electric current iii) power iv) network & v) circuit.

- i) Charge: Charge is an electrical property of the atomic particles of which matter consists, measured in coulombs(C).
- ii) Electric current is the time rate of change of charge, measured in amperes(A). $i = dq/dt$
A direct current (DC) is a current that remains constant with time.
An alternating current (AC) is a current that varies sinusoidally with time
- iii) Power is the time rate of expending or absorbing energy, measured in watts(w).
 $p = dw/dt$ p- Power in watts(w); w- energy in joules (J); t - time in seconds (S); (or) $p = v i$, v - Voltage in volts(V); i – current in amperes(A);
- iv) Network: The inter connection of two or more simple circuit elements forms an electrical network .
- v) Circuit : If the network contains at least one closed path, it is an electric circuit.

6. Define: Node (OR) Junction, Tree & branch

A Node is a point in the network where two or more circuit elements are connected.

Tree A Tree is a complete path including all the nodes.

Branch A branch is a part of the circuit which lies between two junction points.

7. Write down the expression of equivalent resistance for 'n' – number of resistors in series connection.

For 'n' resistors connected in series, the equivalent resistance is given by,
 $R_{eq} = R_1 + R_2 + R_3 + \dots + R_n$

8. Write the Algorithm for Nodal Analysis.

Select a node as the reference node. Assign voltages $V_1, V_2, \dots, V_{n-1}$ to the remaining n-1 nodes. Apply KCL to each of the n-1 nodes. Solve the resulting simultaneous equations to obtain the unknown node voltages.

9. Write the Algorithm for Mesh Analysis. (DEC, '12)

Assign mesh currents $i_1, i_2, \dots, i_n$ to the n meshes. Apply KVL to each of the n meshes. Solve the resulting n simultaneous equations to get the mesh currents.

10. Distinguish between a Loop & Mesh of a circuit (DEC, '10)

The closed path of a network is called a Loop. An elementary form of a loop which cannot be further divided is called a mesh. In other words Mesh is closed path does not contain an other loop within it.

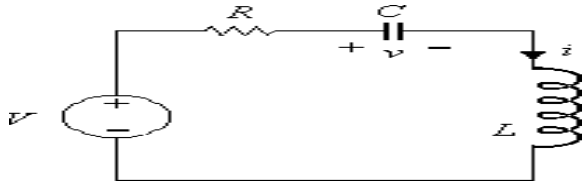
11. State superposition theorem.

It states that the response of a linear circuit with multiple sources is given by algebraic sum of response due to individual sources acting alone.

12. State the steps to solve the super position theorem.

1. Take only one independent voltage or current source.
2. Obtain the branch currents.
3. Repeat the above for other sources.
4. To determine the net branch current just add the current obtained above.

10. Assignments

S.No	Question	BL	CO
1	Define Ohms law, KCL and KVL with examples	1	1
2	A series circuit having pure resistance of $40\ \Omega$, pure inductance of 50.07mH and a capacitance is connected across a 400V, 50Hz Ac supply. This R, L, C combination draws a current of 10A. Calculate (JUN-09)[AU-DEC-10] 1. Power factor of circuit 2. Capacitor value 	2	4
3	Define super position theorem and explain the procedure with one example.	2	3
4	Draw the phasor diagram for a series RLC circuit energized by a sinusoidal voltage showing the relative position of current, component voltage and applied voltage for the following case a) When $X_L > X_C$ b) When $X_L < X_C$ c) When $X_L = X_C$	2	4

11. Part A- Question & Answers

S.No	Question & Answers	BL	CO
1	Define Impedance. Impedance is defined as the opposition of circuit to flow of alternating current. It is denoted by Z and its unit is ohms	1	1
2	State Kirchoff's Voltage law. KVL (Kirchoff's Voltage Law) states that the algebraic sum of all voltages around a closed path (or loop) is zero. (or) Sum of voltage drop = Sum of voltage rise.	1	1
3	Define Real power. The actual power consumed in an AC circuits is called real power. And $P = EI \cos\theta$	1	4
4	Define Reactive power. The power consumed by the pure reactance (X_L or X_C) in an AC circuit is called reactive power. The unit is VAR. and $Q = EI \sin\theta$	1	4

5	Define Apparent power and Power factor. The Apparent power (in VA) is the product of the rms values of voltage and current. $S = V_{rms} I_{rms}$ The Power factor is the cosine of the phase difference between voltage and current. It is also the cosine of the load impedance. And Power factor = $\cos \phi$ The pf is lagging if the current lags voltage (inductive load) and is leading when the current leads voltage (capacitive load).	1	4
6	What is meant by Complex power? Complex power (in VA) is the product of the rms voltage phasor and the complex conjugate of the rms current phasor. As a complex quantity, its real part is real power, P and its imaginary part is reactive power, Q. and $S = P + jQ$	1	4
7	Define Phasor and Phase angle. A sinusoidal wave form can be represented or in terms of a Phasor. A Phasor is a vector with definite magnitude and direction. From the Phasor the sinusoidal wave form can be reconstructed. Phase angle is the angular measurement that specifies the position of the alternating quantity relative to a reference	2	3
8	Write the Algorithm for Nodal Analysis. Select a node as the reference node. Assign voltages $V_1, V_2, \dots, V_{n-1}$ to the remaining n-1 nodes. Apply KCL to each of the n-1 nodes. Solve the resulting simultaneous equations to obtain the unknown node voltages.	1	3
9	Distinguish between a Loop & Mesh of a circuit (DEC, '10) The closed path of a network is called a Loop. An elementary form of a loop which cannot be further divided is called a mesh. In other words Mesh is closed path does not contain an other loop within it.	1	3
10	Write down the expression of equivalent resistance for 'n' – number of resistors in series connection. For 'n' resistors connected in series, the equivalent resistance is given by, $R_{eq} = R_1 + R_2 + R_3 + \dots + R_n$	1	2

12. Part B- Questions

S.No	Question	BL	CO
1	Define Ohms law, KCL and KVL with examples	1	1
2	Define super position theorem and explain the procedure with one example.	2	3
3	Draw the phasor diagram for a series RLC circuit energized by a sinusoidal voltage showing the relative position of current, component voltage and applied voltage for the following case a) When $X_L > X_C$ b) When $X_L < X_C$ c) When $X_L = X_C$	3	4
4	Define the terms form factor, Time period, Active power, Power factor, Phase difference and Inductive reactance	2	4
5	Explain the relation between voltage and current when single phase AC Supply given to Pure RLC Elements	3	4

13. Supportive Online Certification Courses

1. Basic Electrical Circuits By Prof. Ankush Sharma, conducted by IIT Kanpur – 4 weeks
2. Basic Electrical Engineering By Prof. Bhattacharya, conducted by IIT Madras – 08 weeks.

14. Real Time Applications

S.No	Application	CO
1	In electrical engineering, single-phase electric power is the distribution of alternating current electric power using a system in which all the voltages of the supply vary in unison. Single-phase distribution is used when loads are mostly lighting and heating, with few large electric motors.	4
2	Single phase supply is used for small electrical appliances and equipment, mostly for home applications.	4
3	All 230 V 50 Hz supply is used in electrical appliances like TV, EV charger level 2, fan, tubelight, LED bulb etc.	4
4	In India single phase is used for converting electrical energy into other useful forms like rotation (motion), light, heat, sound etc. Usually this is 3 to 5 KW.	4

15. Contents Beyond the Syllabus

1. Graphical evaluation of frequency response
2. Economics of power supply system

16. Prescribed Text Books & Reference Books

Text Books:

1. D. P. Kothari and I. J. Nagrath - "Basic Electrical Engineering" - Tata McGraw Hill -2010.
2. V.K. Mehta & Rohit Mehta, "Principles of Power System" – S.Chand – 2018.

References:

1. L. S. Bobrow - "Fundamentals of Electrical Engineering" - Oxford University Press -2011.
2. E. Hughes - "Electrical and Electronics Technology" - Pearson - 2010.
3. C.L. Wadhwa – "Generation Distribution and Utilization of Electrical Energy", 3rd Edition, New Age International Publications.

17. Mini Project Suggestion

1. Temperature Controlled System:

This project shows a temperature-controlled system. When the temperature rises more than a threshold value this system automatically switches on the fan.

2. Conversion of Single Phase to Three Phase Supply:

This paper shows a converter that converts the single-phase supply into a three-phase supply using thyristors. Go through the paper for more information.

3. Smoke Detector Alarm Circuit:

Here is the circuit showing a smoke detector alarm. This can also be used to indicate the fire. This circuit uses a smoke detector and an LM358 Comparator.

4. Railway Security System based on Wireless Sensor Networks:

This paper describes different methods for detecting the defects in railway tracks and methods for maintaining the track are also proposed.

5. Control Electrical Devices From your android phone:

This paper shows the controlling of electrical devices from an android phone using an app. This project uses Arduino for controlling the devices.